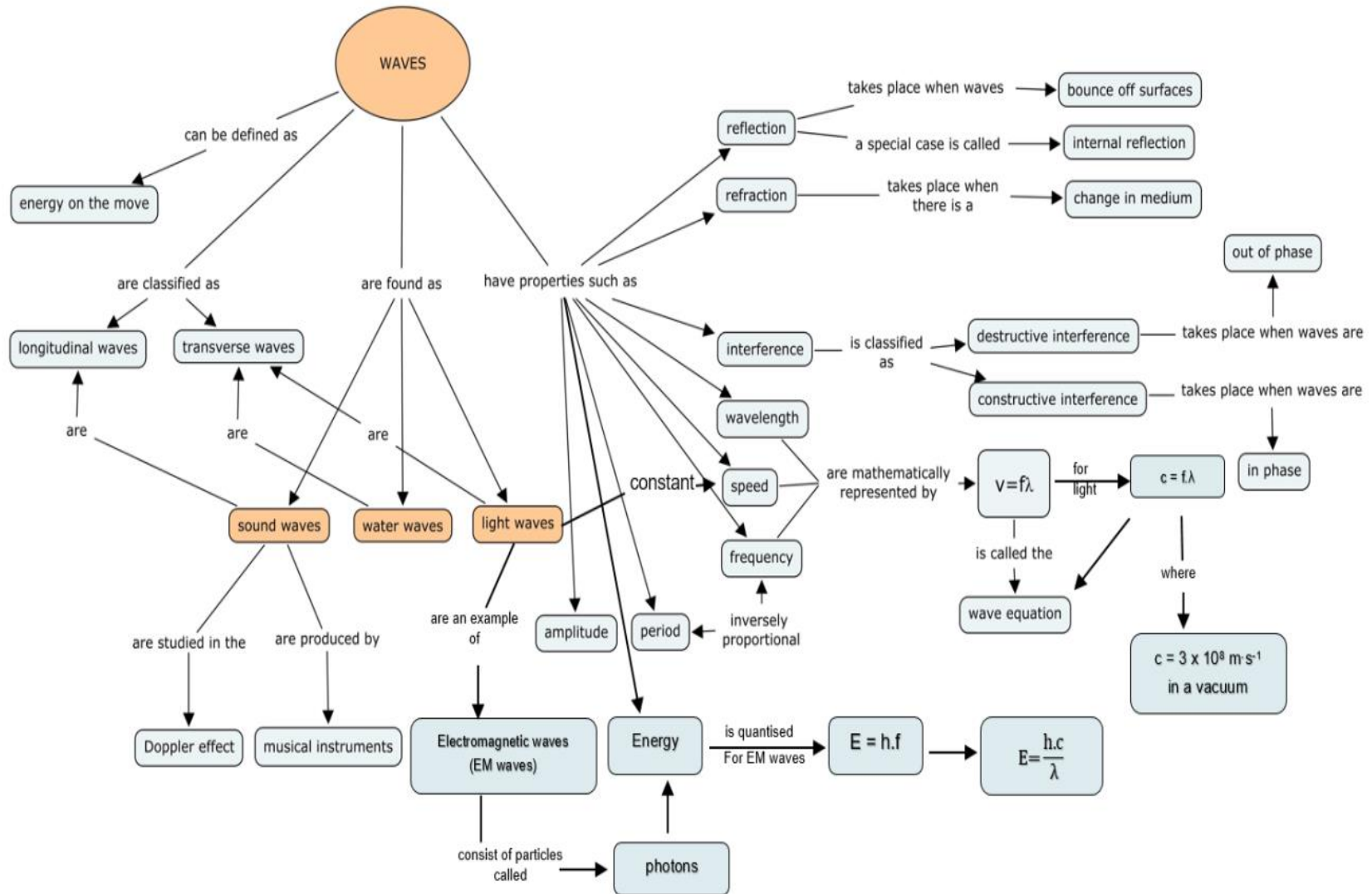


Concept Map of Waves, Light & Sound

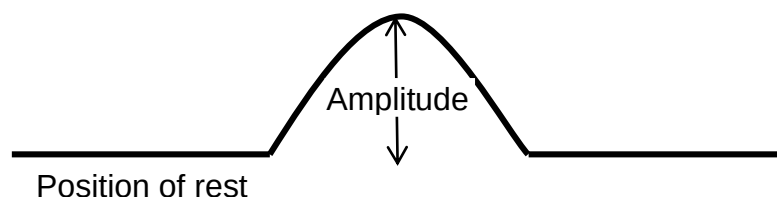


TRANSVERSE PULSES ON A STRING OR SPRING**PULSE AND AMPLITUDE**

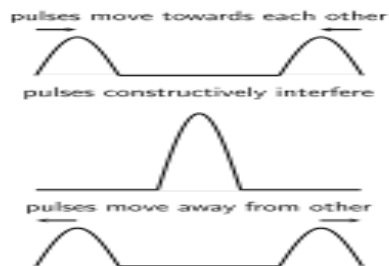
- ✓ A **pulse** is a single disturbance (vibration) in a medium.
- ✓ A **transverse pulse** is a pulse in which particles of the medium moves at right angles to the direction of motion of the pulse.
- ✓ A transverse pulse has amplitude and pulse length.
- ✓ **Amplitude** is a maximum disturbance of a particle from its rest (equilibrium) position.

Practical activity: Generating a pulse

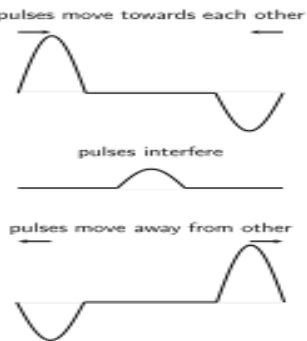
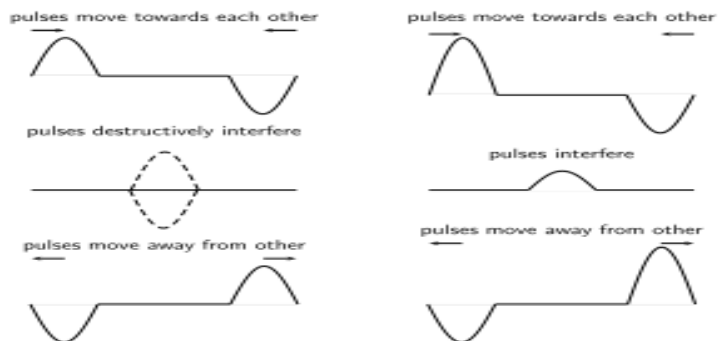
- ✓ Flick the rope or slinky spring at one end only once, while the other end is fixed.
- ✓ Learners observe the motion the pulse

**SUPERPOSITION OF PULSES**

- ✓ **The principle of superposition** is the addition of the amplitudes of two pulses that occupy the same space at the same time.
- ✓ **Constructive interference** takes place when two pulses meet each other to create a larger pulse (see illustration below) i.e. when a crest meets another crest or a trough meets another trough.



- ✓ **Destructive interference** takes place when two pulses meet and results in a pulse of reduced amplitude i.e. when a trough meets a crest.
- ✓ The **amplitude** of the resulting pulse is the sum of the amplitudes of the two initial pulses, but one of the amplitudes will be a negative number.



TRANSVERSE WAVES

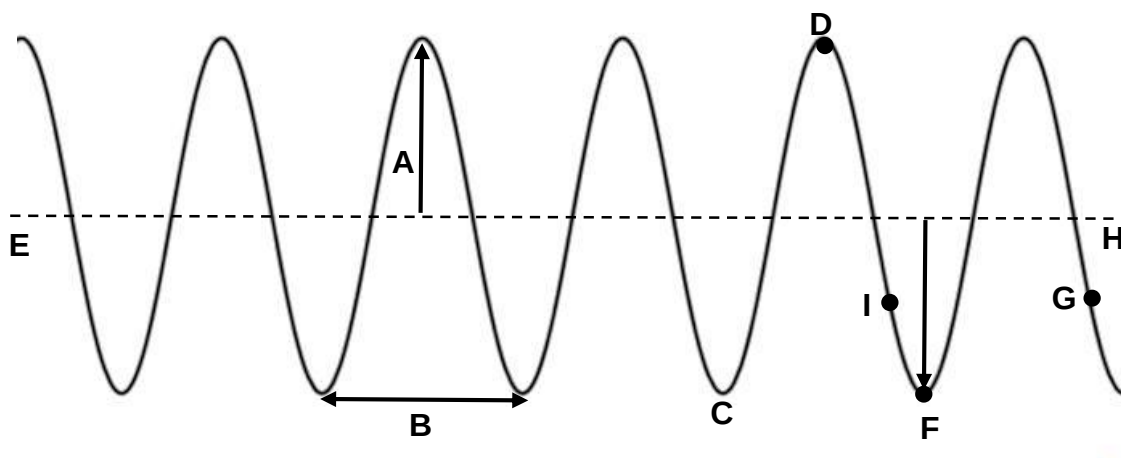
- ✓ A **transverse wave** is the wave in which the particles of the medium vibrate at right angles to the direction of motion of the wave.

Practical activity: generating a transverse wave

- ✓ Flip one end of the rope or slinky spring up and down continuously to create a train of pulses.

DEFINITIONS

Diagram of a transverse wave



A: Amplitude

- ✓ Amplitude is the maximum displacement of a particle from its equilibrium position (rest position).

B. Wavelength

- ✓ Wavelength is the distance between two successive points in phase.

C: Trough

- ✓ Trough is the lowest point on a wave.

D: Crest

- ✓ Crest is the highest point or peak on a wave.
- ✓ **EH: Equilibrium** position/position of rest is the centre line that shows the position of undisturbed string or spring.
- ✓ **Points in phase: C and F are in phase G and I are in phase**
- ✓ Points in phase are two points in phase separated by a whole number multiple of complete wavelengths.
- ✓ **F and G: Points out of phase**
- ✓ **Points out of phase** are points that are not separated by a whole number multiple of complete wavelengths.

- ✓ **Period (T)** is the time taken for one complete wave pulse.

$$\circ T = \frac{1}{f}$$

- Period is measured in seconds(s).

- ✓ **Frequency (f)** is the number of wave pulses per second.

$$\circ f = \frac{1}{T}$$

- ✓ Frequency is measured in hertz (Hz).
- ✓ **Wave speed (v)** is the distance travelled by a point on a wave per unit time
- ✓ $v = f \cdot \lambda$ (wave equation)

Quantity	Symbol	Unit	Symbol
Speed of wave	v	metre per second	$\text{m} \cdot \text{s}^{-1}$
Frequency	f	Hertz	Hz
Wavelength	λ	Metre	m

Worked Examples**Example 1**

When a string is vibrated at a frequency of 10 Hz, a transverse wave of wavelength of 0,5m is produced. Determine the speed of the wave as it travels along the string.

Solution

Formula:	$v = f \cdot \lambda$
Substitution:	$= (10)(0,5)$
Answer:	$= 5 \text{ m} \cdot \text{s}^{-1}$

Example 2

A wave with a wavelength of 1,5 m moves past a point in 20 seconds. 60 complete waves are completed. Calculate the:

2.1 frequency of the waves

2.2 period of the waves

2.3 speed of the waves

SOLUTIONS

2.1 $f = \frac{\text{number of waves}}{\text{time}}$

$$= 60 \div 20$$

$$= 3 \text{ Hz}$$

2.2 $T = 1 \div f$

$$= 1 \div 3$$

$$= 0,33 \text{ s}$$

2.3 $v = f \cdot \lambda$

$$= (3)(1,5)$$

$$= 4,5 \text{ m} \cdot \text{s}^{-1}$$

EXAMPLE 3

A wave has a frequency of 400Hz and a wavelength of 0,75 m. Calculate the speed of the wave.

SOLUTION

Given data:

$$f = 400 \text{ Hz}$$

$$\lambda = 0,75 \text{ m}$$

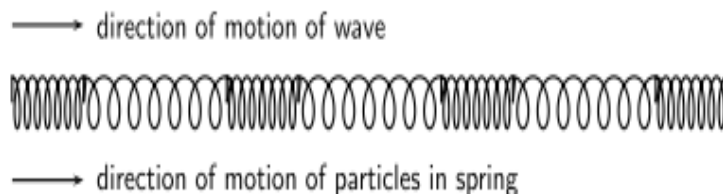
$$\begin{aligned} v &= f \cdot \lambda \\ &= (400)(0,75) \\ &= 300 \text{ m.s}^{-1} \end{aligned}$$

LONGITUDINAL WAVES

- ✓ A longitudinal wave is a wave where the particles of the medium vibrate parallel to the direction of motion of the wave.

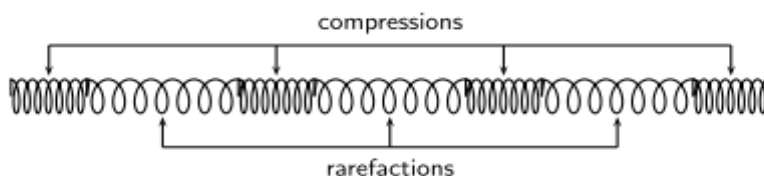
Practical Activity**How to generate a longitudinal wave**

- ✓ Take a slinky spring and hang it from the ceiling. Pull the free end of the spring and release it.

Diagram for longitudinal waves

Properties of a longitudinal wave

- ✓ A **compression** is a region of high pressure in a longitudinal wave.
- ✓ A **rarefaction** is a region of low pressure in a longitudinal wave.



- ✓ The wave equation $v = f\lambda$ is also used to solve problems involving longitudinal waves.

SOUND WAVES

- ✓ A sound wave is an example of a longitudinal wave
- ✓ Vibrations in a medium in the direction of propagation create sound waves.
- ✓ Sound wave cannot be propagated in a vacuum.
- ✓ Speed of sound depends on the properties of a medium; it travels faster in a medium with particles closely packed.
- ✓ Sound waves travel faster through liquids, like water, than through the air because water is denser than air.
- ✓ Sound waves travel faster in solids than in liquid because a solid is denser than a liquid
- ✓ e.g.

Medium	Speed (m.s^{-1})
Air	340
Water	1500
Steel	5900

ECHOES

Sound waves can be reflected by hard surfaces like walls and cliffs. The reflected sound is called an echo.

CALCULATIONS BASED ON ECHOES**Example 1**

A man stands between two tall buildings. When he claps his hands, he hears an echo from one building after two seconds and an echo from the other building after three seconds. Calculate the distance that the buildings are apart from each other. Take the speed of sound as 330 m.s^{-1} .

Solution.

For building 1:

$$\begin{aligned}\text{Distance} &= \text{speed} \times \text{time} \\ &= 330 \times 1 \\ &= 330 \text{ m.}\end{aligned}$$

For building 2:

$$\begin{aligned}\text{Distance} &= \text{speed} \times \text{time} \\ &= 330 \times 1,5 \\ &= 495 \text{ m.}\end{aligned}$$

$$\begin{aligned}\text{Distance between 2 buildings} &= 330 + 495 \\ &= 825 \text{ m.}\end{aligned}$$

Example 2

The speed of sound in seawater is 1500 m.s^{-1} . A fishing boat sends signals out that are received after 20s. Calculate the depth of the seabed at that position.

Solution.

(time taken for signal to reach the seabed = $20/2 = 10 \text{ s}$)

$$\begin{aligned}\text{Distance} &= \text{speed} \times \text{time} \\ &= 1500 \times 10 \\ &= 15\,000 \text{ m.}\end{aligned}$$

CHARACTERISTICS OF SOUND**PITCH**

- ✓ The frequency of a sound wave is what your ear understands as pitch.
- ✓ The pitch represents how high or how low the note sounds.
- ✓ A higher frequency sound has a higher pitch, and a lower frequency sound has a lower pitch.(frequency is directly proportional to pitch)

LOUDNESS

- ✓ The amplitude of a sound wave determines its loudness or volume.
Loudness is the listener's evaluation of amplitude.
- ✓ A larger amplitude means a louder sound, and a smaller amplitude means a softer sound. (loudness is directly proportional to the amplitude)
- ✓ Loudness is measured in decibels (dB)

tone

- ✓ The tone is a measure of the quality of the sound wave
- ✓ Sound produced by a drum is not of quality because it has different frequencies and it lasts for a short while.(this is noise)

Example.

Two musical notes have the same amplitude, but note A has twice the frequency of note B. In which respects would:

- a) their sounds differ; and
- b) their sounds be the same?

Solution

- a) Pitch
- b) Loudness

ULTRASOUND

- ✓ Ultrasound is sound with a frequency that is **higher than 20 kHz**.
- ✓ The most common use of ultrasound is **to create images**.
- ✓ The use of ultrasound to create images is based on the **reflection and transmission** of a wave at a boundary.
- ✓ When an ultrasound wave travels inside an object that is made up of different materials, part of the wave is reflected and part of it is transmitted.
- ✓ The reflected rays are detected and used to construct an image of the object.

USES OF ULTRASOUND

In medicine:

- ✓ To visualise muscle and soft tissue, e.g. during pregnancy
- ✓ To generate local heating in biological tissue e.g. to break up kidney stones.

In industry:

Ultrasonic cleaners are used to clean jewellery, lenses and other optical parts, watches, dental instruments, surgical instruments and industrial parts.

ELECTROMAGNETIC RADIATION

- ✓ It is produced by charges accelerating through space.
- ✓ There are both natural and man-made sources of electromagnetic radiation. The sun, for instance, is an intense natural source of radiation that can have both positive and negative effects on living things. The sun produces both visible (light) and invisible (ultra violet-rays) electromagnetic radiation streams.
- ✓ Man-made sources of electromagnetic radiation include X-rays, radio waves, microwaves and gamma rays. Humans to power machines and to increase communication abilities use microwaves and radio waves. Gamma rays are used in chemotherapy. Cell phones, radios, microwave ovens, and radar all create electromagnetic radiation.
- ✓ Either using the wave nature or the particle nature of the radiation (visible light) can explain the properties of electromagnetic radiation.
- ✓ This property of electromagnetic radiation to have a wave nature and a particle nature is known as **wave-particle duality**. Light is an example of wave-particle duality.
- ✓ *Properties of electromagnetic waves*
- ✓ They originate from accelerating electric charges
- ✓ They propagate as electric and magnetic fields that are perpendicular to each other
- ✓ They can travel through a vacuum
- ✓ They travel with a speed of $3 \times 10^8 \text{ m}\cdot\text{s}^{-1}$

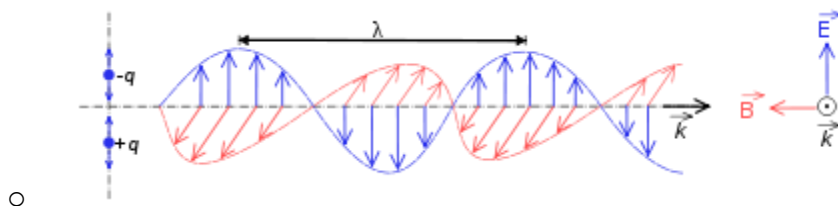
Dual nature of electromagnetic radiation

- ✓ Diffraction and interference show that light has a wave particle
- ✓ The photo electric effect demonstrates the particle nature

Wave nature will be discussed in grade 11 and particle nature will be discussed in grade 12.

Accelerating Charge

- ✓ Accelerating charges emit electromagnetic waves.
- ✓ A changing electric field generates a magnetic field and a changing magnetic field generates an electric field. This is the principle behind the propagation of electromagnetic waves.
- ✓ Electromagnetic waves do not need a medium to travel through.
- ✓ **Propagation of Electromagnetic Radiation**
- ✓ Explanation and diagram of the transmission or propagation of electromagnetic radiation:
- ✓ Electromagnetic (EM) waves propagate when an electric field oscillating in one plane produces a magnetic field oscillating in a plane at right angles to it.
- ✓ The electric field(s) and magnetic field(s) must remain perpendicular to each other.
- ✓ EM wave propagation is a mutual induction because the changing electric field is responsible for inducing the magnetic field and vice versa.
- ✓ **E** is used to show electric field and **B** is used to show magnetic field (refer to the diagram of mutual regeneration of electric field and magnetic field below).



- ✓ These mutually regenerating fields (Electromagnetic radiation) travel through space at a constant speed of $3 \times 10^8 \text{ m.s}^{-1}$. This constant speed of light is represented by c .

The Electromagnetic Spectrum

EM radiation is classified into types according to the frequency and wavelength of the wave. These types of EM radiation in the table below are in order of decreasing frequency and increasing wavelength (from top to bottom).

Category	Range of Wavelength in (nm)	Range of Frequency in (Hz)
Gamma Rays	< 1	$> 3 \times 10^{19}$
X-Rays	$1 - 10$	$3 \times 10^{17} - 3 \times 10^{19}$
Ultraviolet Light	$10 - 400$	$7.5 \times 10^{14} - 3 \times 10^{17}$
Visible Light	$400 - 700$	$4.3 \times 10^{14} - 7.5 \times 10^{14}$
Infrared rays	$700 - 10^5$	$3 \times 10^{12} - 4.3 \times 10^{14}$
Microwaves	$10^5 - 10^8$	$3 \times 10^9 - 3 \times 10^{12}$
Radio Waves	$> 10^8$	$< 3 \times 10^9$

The Electromagnetic Spectrum in order of increasing wavelength

G X U V I M R

The Electromagnetic Spectrum in order of decreasing wavelength

R M I V U X G

Conversions Table

Unit	Symbol	Unit in metres
Micrometres	μm	10^{-6}
Nanometres	nm	10^{-9}
Pico metres	pm	10^{-12}

EM radiation is a wave and this equation also applies: $v = f \lambda$, however v is replaced by c and the wave equation becomes $c = f \lambda$

Calculations using light wave equation $c = f \lambda$

Worked examples

1. Calculate the frequency of an EM wave with a wavelength of 200nm.

Note: nano(n) = 10^{-9} .

Solution:

Step 1: What do we know?

We can identify two properties of the radiation

$$\lambda = 200 \text{ nm} \quad c = 3 \times 10^8 \text{ m.s}^{-1} \quad f = ?$$

Step 2: Apply the wave equation

$$c = f \lambda$$

$$3 \times 10^8 = f \times 200 \times 10^{-9}$$

$$= 1,5 \times 10^{15} \text{ Hz.}$$

2. Calculate the wavelength of an EM wave with $6 \times 10^{-19} \text{ J}$ of energy.

$$c = 3 \times 10^8 \text{ m.s}^{-1} \quad h = 6,63 \times 10^{-34} \text{ J.s} \quad \lambda = ?$$

$$E = hf = (hc) \div \lambda$$

$$6 \times 10^{-19} = (6,63 \times 10^{-34} \times 3 \times 10^8) \div \lambda$$

$$\lambda = (6,63 \times 10^{-34} \times 3 \times 10^8) \div 6 \times 10^{-19}$$

$$= 3,32 \times 10^{-7} \text{ m}$$

$$= 332 \times 10^{-9} \text{ m}$$

$$= 332 \text{ nm}$$

Some Uses of EM Radiation

The table below shows examples of some uses of electromagnetic waves.

Category	Uses
Gamma rays	Used to kill the bacteria in marshmallows and to sterilise medical equipment
X – rays	Used to image bone structures
Ultraviolet light	Bees can see into the ultraviolet because flowers stand out more clearly at this frequency
Visible light	Used by humans to observe the world
Infrared	Night vision, heat sensors, laser metal cutting
Microwave	Microwave ovens, radar
Radio waves	Radio, television broadcasts

Penetrating Ability of EM Radiation

EM radiation has different degrees of penetration because of different frequencies they possess. In a human body EM radiation penetrates as follows:

- ✓ Visible light is reflected off the surface of the human body.
- ✓ Ultra-violet light (from sunlight) damages the skin.
- ✓ X-rays penetrate the skin and bone
 - X-rays have a higher penetrating ability than visible light and therefore X-rays have a much higher frequency (energy) than visible light.
- ✓ So, EM with higher frequency has higher degree of penetration than those with low frequency.

- ✓ Gamma rays, X-rays and Ultra violet rays are very dangerous EM rays. Their radiation is called ionising radiation.
- ✓ Ionising radiation is the transfer of energy when EM passes through matter, breaking molecular bonds and creating ions.
- ✓ Excessive exposure to radiation, including sunlight, X-rays and all nuclear radiation can cause destruction of biological tissue.

Ultraviolet(UV) Radiation and the skin:

UV has different ranges of frequency namely the UVA and UVB, which affect the skin as follows:

UVA light

- ✓ Speeds up ageing of skin
- ✓ Damages DNA
- ✓ Can possibly lead to skin cancer
- ✓ Does not cause sunburn but penetrates deeply in the skin.

○ UVB light

- ✓ can cause skin cancer
- ✓ excites DNA leading to cancerous mutations

Discussion of Radiation from Cell Phones

Cell Phone Electromagnetic Radiation

- ✓ It is commonly agreed that there are low-levels of **radiation** emitted from **cell phones**, with some emitting more than others.
- ✓ The exact source of **radiation** in a **cell phone** is from the transmitter, a device located near the antenna that converts audio data into **electromagnetic** waves. The amount of **radiation** a **cell phone** can emit is limited by legal restrictions.
 - The best method of cell phone radiation protection is to keep the phone as far from the body as possible while it is turned on. Keeping calls short as well as making them from good reception areas will also reduce exposure.

Cell Phone Radiation Danger:

- ✓ The most important time for **cell phone radiation** protection is during an actual **phone** call when a **phone** emits the most **radiation**.
- ✓ An increased probability of being involved within a car accident while using a cell phone is one of the most common cell phone risks.
- ✓ Many times, a cell phone user will touch the phone with his or her hands, cheeks, mouth, and ears. Cell phones are made of plastic, metals, and other materials that could potentially cause allergic reactions to nickel. This reaction is known as **contact dermatitis**. Most times, the symptoms of this type of allergic reaction include redness and/or blisters.

Photons

- ✓ A photon is a quantum (energy packet) of light. A particle of light is called a photon.
- ✓ The energy of a photon can be calculated using the formula: **$E = hf$** or
 - **$E = hc/\lambda$**
- ✓ [From the EM radiation wave equation **$c = f \cdot \lambda$** , rearranging **f** leads to **$f = c/\lambda$** then substitute into the energy of a photon equation **$E = hf$** to get **$E = (hc)/\lambda$**]
- ✓ **E** is the energy of the photon that is calculated in joules (J), **f** is the frequency that is calculated in hertz (Hz) and **h** is Plank's constant where
 - **$h = 6,63 \times 10^{-34}$ Js**. The higher the frequency of EM radiation, the higher the energy.

Calculating the energy of a photon using the equation: $E = hf$

Worked example

1. Calculate the energy of a photon with a frequency of 3×10^{18} Hz

Solution:

Step1: Analyse the question

Here you are asked to determine the energy of a photon given the frequency (f).

Given: $f = 3 \times 10^{18}$ Hz $h = 6.63 \times 10^{-34}$ J.s

Step 2: Apply the equation for the energy of a photon

$$\begin{aligned} E &= hf \\ &= 6,63 \times 10^{-34} \times 3 \times 10^{18} \\ &= 1,99 \times 10^{-25} \text{ J} \end{aligned}$$

2. Calculate the energy of a photon with a frequency of $4,3 \times 10^{14}$ Hz

Calculating using energy of a photon equation: $E = hc \div \lambda$

Worked example

1. Determine the energy of a photon of EM radiation with a wavelength of 600 nm.

Solution:

Step 1: Analyse the question

Here you are asked to determine the energy of a photon given the wavelength.

Given: $\lambda = 600$ nm and $h = 6.63 \times 10^{-34}$ J.s

Step 2: Apply principles

Firstly we determine the frequency from the equation $c = f \cdot \lambda$

Then rearrange: $f = c \div \lambda$

Then substitute f into energy of a photon equation $E = hf$

$$\begin{aligned} E &= hc \div \lambda \\ &= (6,63 \times 10^{-34}) \times 3 \times 10^8 \div 600 \times 10^{-9} \end{aligned}$$

$$= 9.945 \times 10^{-19} \text{ J}$$

2. Determine the energy of a photon of electromagnetic radiation with a wavelength of $4.2 \times 10^{-7} \text{ m}$.

$$E = hc \div \lambda$$

$$= (6.63 \times 10^{-34} \times 3 \times 10^8) \div 4.2 \times 10^{-7}$$

$$= 4.74 \times 10^{-19} \text{ J}$$

Waves, legends and folklores

Waves associated with natural disasters:

Natural disasters are: earthquakes, tsunamis and floods

- ✓ Seismic waves are associated with earthquakes.
- ✓ Ocean waves are associated with tsunamis.
- ✓ Heat wave is associated with excessively hot weather and floods (inland and coastal) e.g. heat wave El Nino in 1998.
- ✓ Sea waves called tsunamis is also associated with coastal flooding.